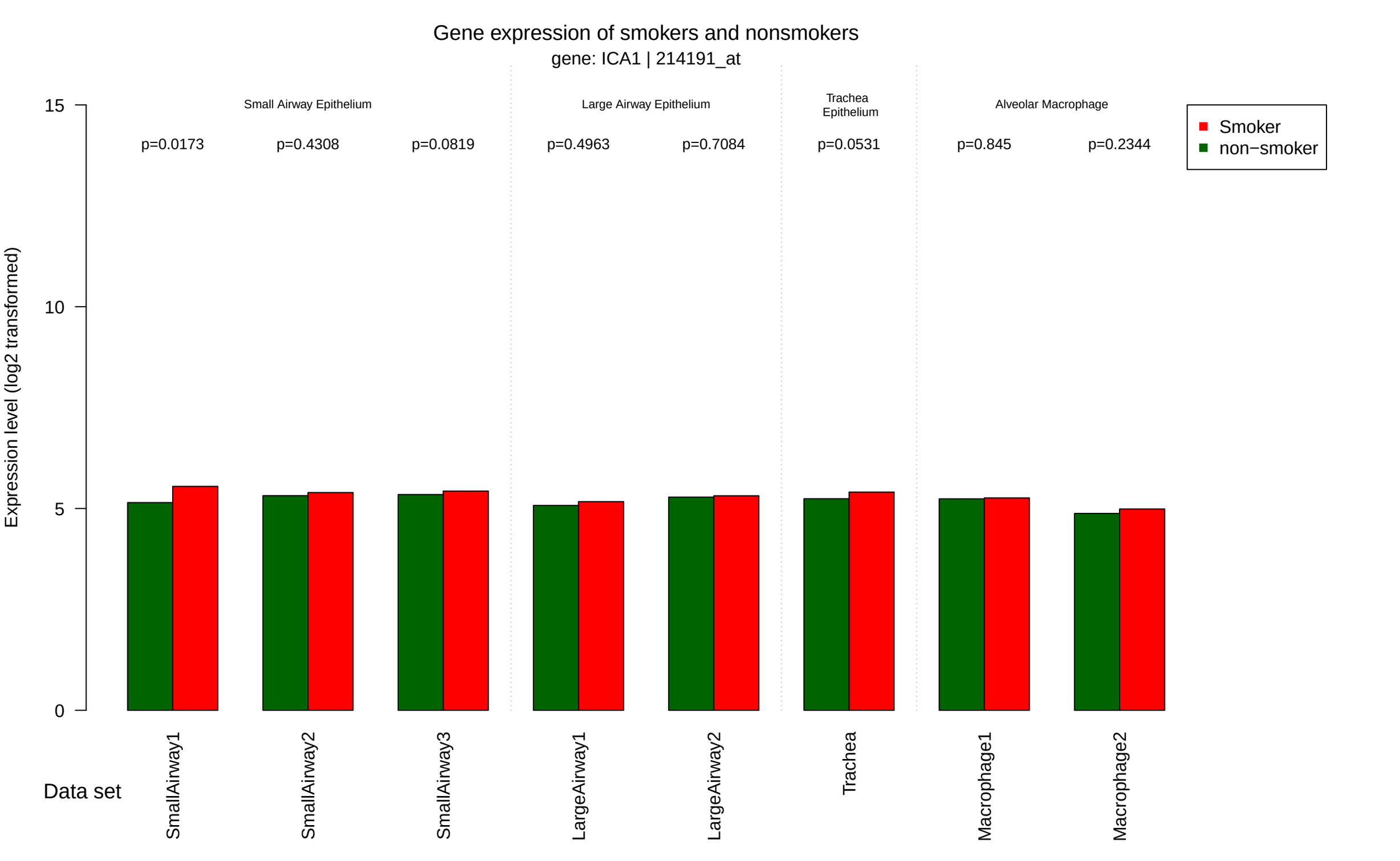
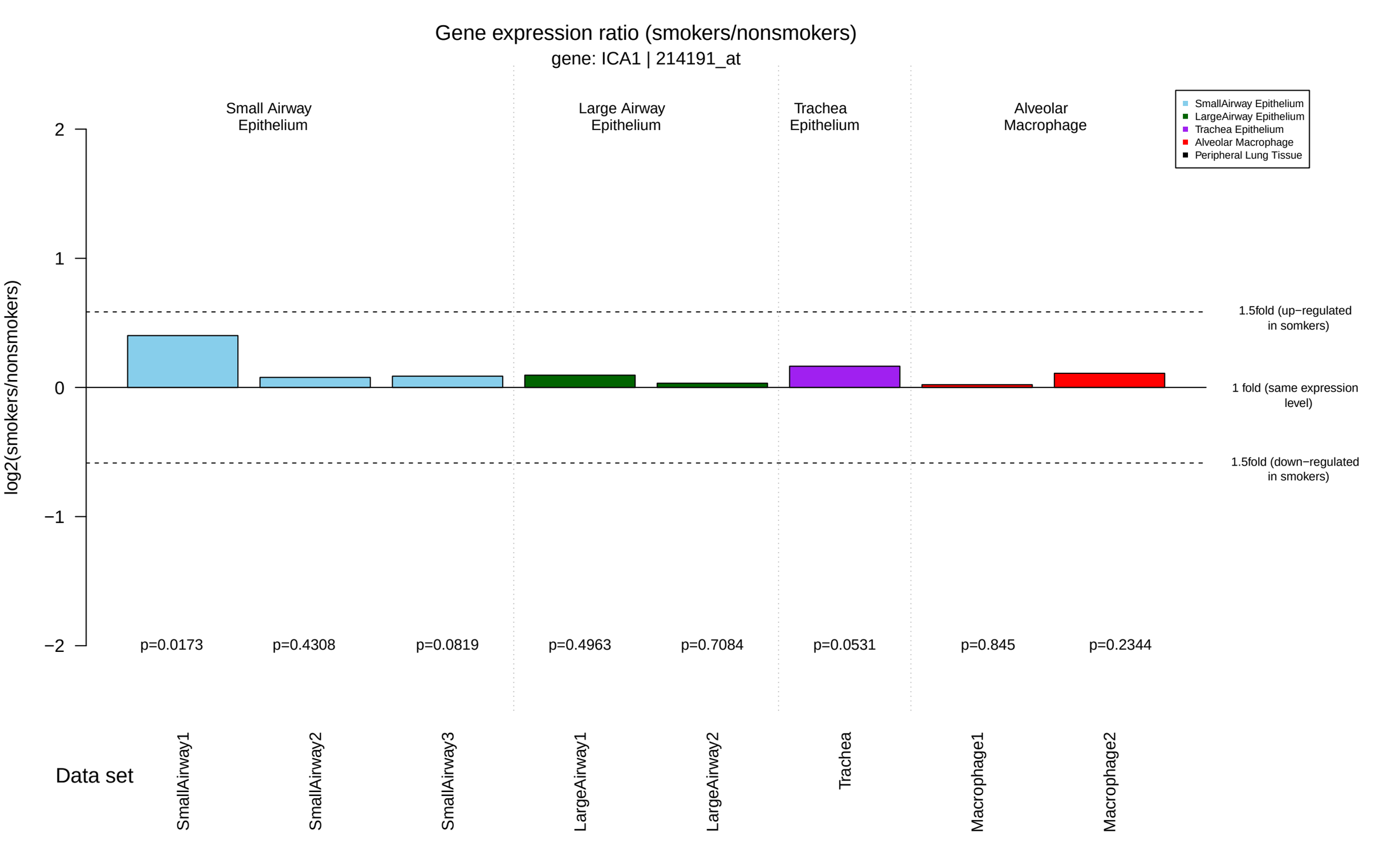
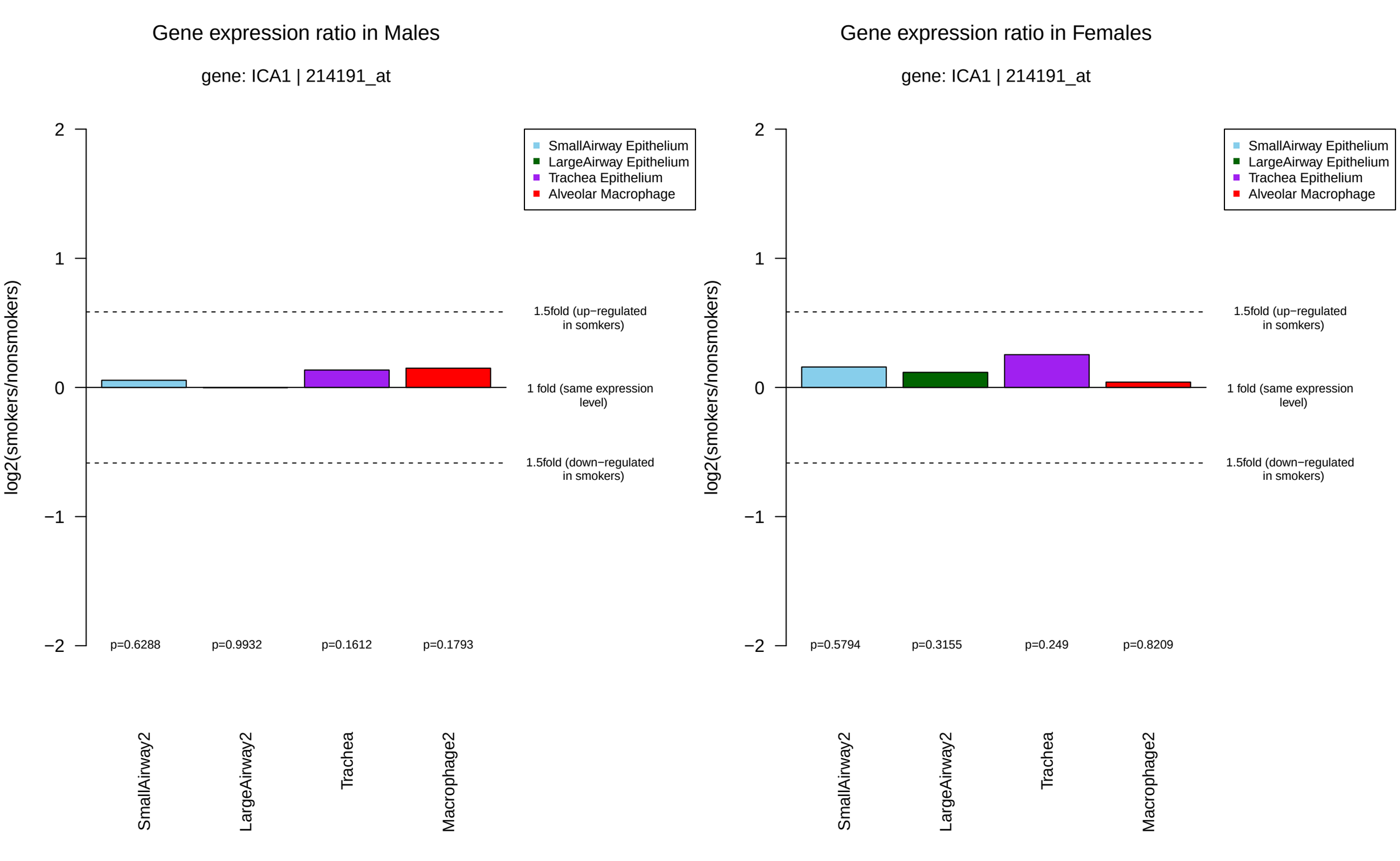
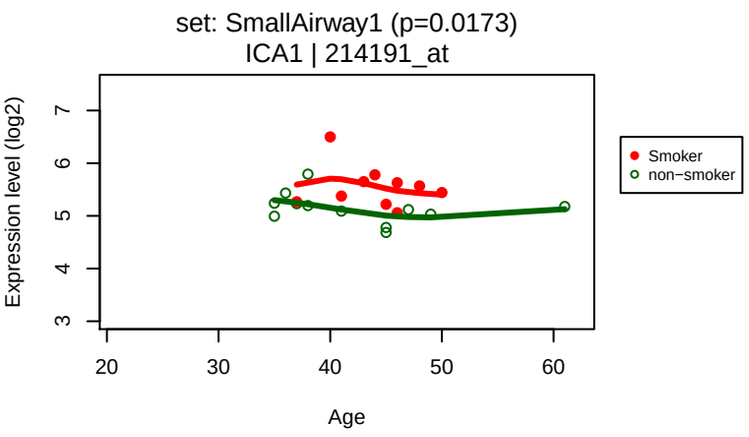
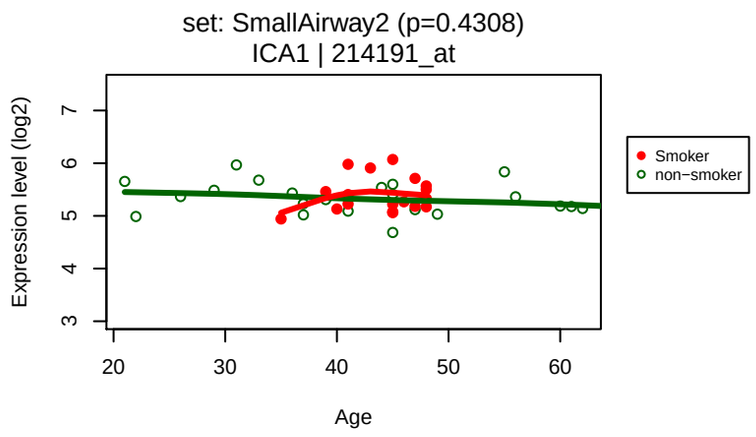
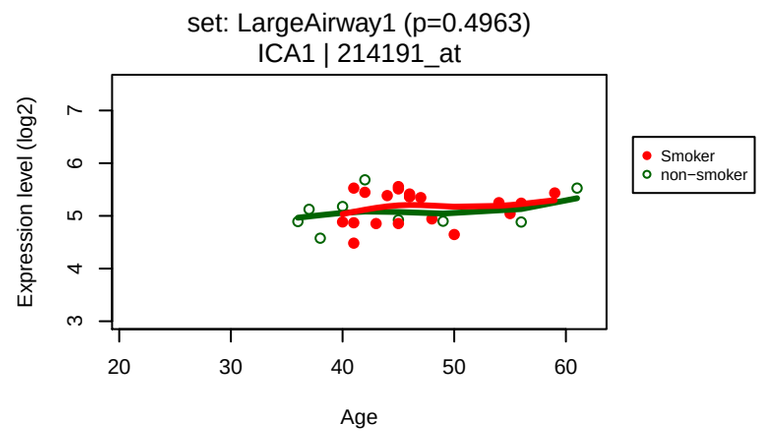

Gene expression of smokers and nonsmokers
gene: ICA1 | 214191_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0173
p=0.4308
p=0.0819
p=0.4963
p=0.7084
p=0.0531
p=0.845
p=0.2344
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

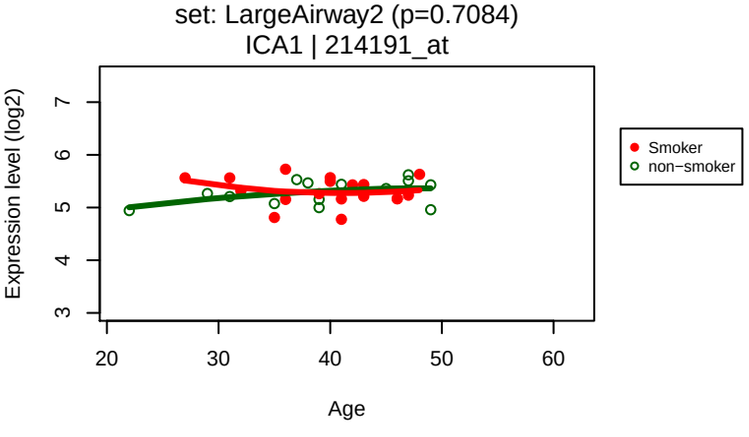

Gene expression ratio (smokers/nonsmokers)
gene: ICA1 | 214191_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0173
p=0.4308
p=0.0819
p=0.4963
p=0.7084
p=0.0531
p=0.845
p=0.2344
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

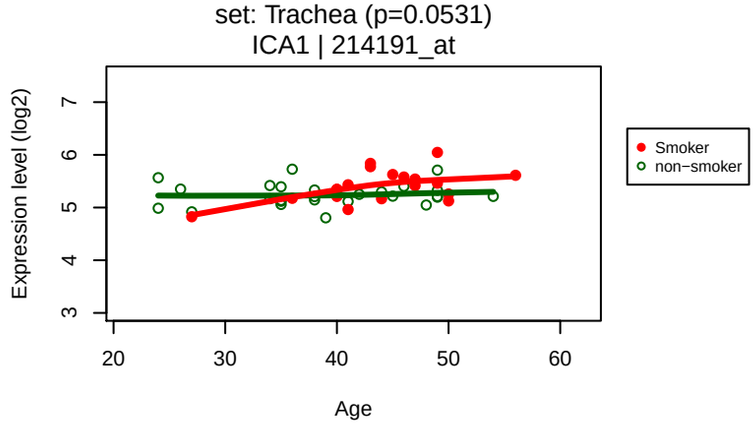

Gene expression ratio in Males
gene: ICA1 | 214191_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6288
p=0.9932
p=0.1612
p=0.1793
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ICA1 | 214191_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5794
p=0.3155
p=0.249
p=0.8209
SmallAirway2
LargeAirway2
Trachea
Macrophage2

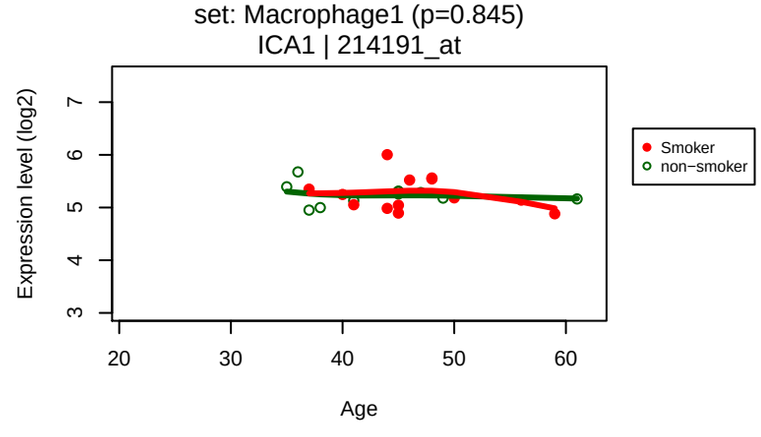

set: SmallAirway1 (p=0.0173)
ICA1 | 214191_at
Expression level (log2)
Age
Smoker
non−smoker

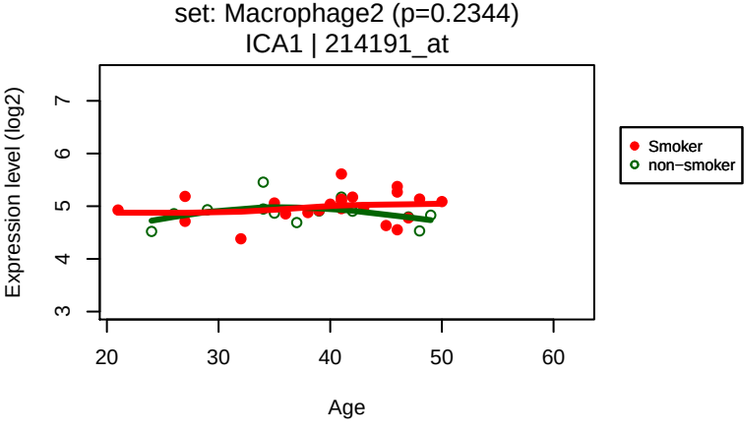

set: SmallAirway2 (p=0.4308)
ICA1 | 214191_at
Expression level (log2)
Age
Smoker
non−smoker

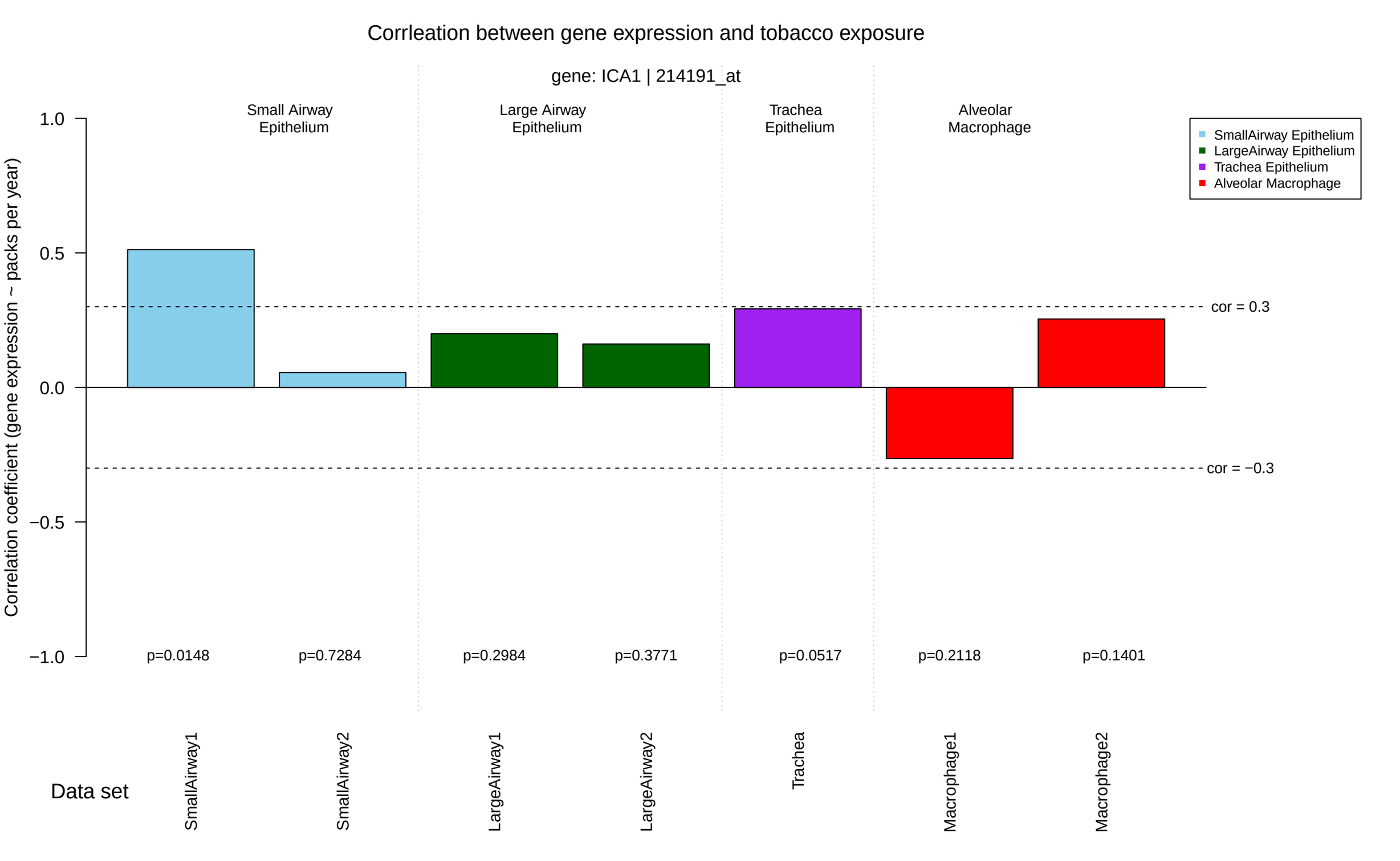

set: LargeAirway1 (p=0.4963)
ICA1 | 214191_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7084)
ICA1 | 214191_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0531)
ICA1 | 214191_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.845)
ICA1 | 214191_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2344)
ICA1 | 214191_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ICA1 | 214191_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0148
p=0.7284
p=0.2984
p=0.3771
p=0.0517
p=0.2118
p=0.1401
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2